MIL-P-52455A(ME)
23 January 1970
SUPERSEDING
MIL-P-52455(MO)
12 November 1965

MILITARY SPECIFICATION

PIN REGISTER PUNCH BOARD,

PIN REGISTER BARS, AND PIN REGISTER PLATES

1. SCOPE

1.1 This specification covers a hand-operated, pin register punch board hereinafter called "pin register board", pin register bars, and pin register plates for accurately alining photolithographic color separation materials.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Military

MIL-P-21105	- Plastic, Sheet, Acrylic, Heat Resistant, Utility Grade.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.

FSC 6675

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A276 - Stainless and Heat-Resisting Steel Bars and Shapes.
B221 - Aluminum-Alloy Extruded Bars, Rods, Shapes, and Tubes.
D702 - Cast Methycrylate Plastic Sheets, Rods, Tubes, and Shapes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

AMERICAN IRON AND STEEL INSTITUTE

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, N. Y. 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook - J916 - Recommended Practices.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N. Y. 10001.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, D. C. 20036.)

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Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The pin register board shall consist of precision punches mounted on a metal bar or frame with a 28- by 32-inch translucent plastic sheet. Pin register bars shall be strips of metal with the specified number and size of studs, buttons, pins, or dowels, hereinafter called "pins", mounted thereon. Pin register plates shall be single studs, pins, dowels, or buttons mounted on metal plates.

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Corrosion-resistant metals shall be used or the metals shall be treated to resist corrosion.

3.2.1 Sheet plastic. Sheet plastic shall conform to MIL-P-21105, transparent or translucent, or ASTM D702, type II, grade 1 or 2, and shall transmit not less than 80 percent of incident light.

3.2.2 Corrosion-resistant steel. Corrosion-resistant steel shall conform to ASTM A276, type 302 except pin register plates which shall conform to ASTM A276, type 303.

3.2.3 Tool steel. Tool steel shall conform to AISI class A2 and O1.

3.2.4 Aluminum. Aluminum, when used, shall conform to ASTM B221, alloy 6063-T5 or 6061-T6511.

3.3 Construction. The pin register board shall not be damaged or distorted by loadings of 94 pounds laterally or 50 pounds vertically. Frames, when used, shall be fabricated of aluminum alloy. Felt or rubber padding around the bottom edge of each board and rubber feet shall prevent the board from marring the surface of a table, such as the light table, while the board is in use. The punches shall be mounted on a metal strip not less than 32 by 2 inches; the strip shall be of not less than 1/8-inch steel, 1/4-inch aluminum alloy, or not less than 1- by 2-inch rectangular, aluminum-alloy tubing with 1/8-inch walls. The punches shall be mounted with their throat openings flush with the working surface so that the materials to be punched will lie flat in one plane. Each punch shall be replaceable with no change in tolerances.

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3.4 Working surface. The working surface of the pin register board shall be a 32- by 28-inch clear or translucent white plastic sheet. An aluminum frame may be placed around the edge of the working surface. When a frame is used, the plastic working surface shall be not less than 1/4 inch in thickness. When a frame is not used, the plastic working surface shall be not less than 3/8 inch thick. There shall be a clear viewing area of not less than 29-7/8 by 24-3/4 inches, unobstructed by the punches, padding, or framing. The working surface shall be flat to within 0.05 inch, both before and after the loading specified in 3.3. Fasteners shall be countersunk. A centerline shall be scribed 0.005 inch deep and 0.005 inch wide, both plus 0.002 minus 0.001 inch, equidistant between the two outside punches and at 90 degrees plus or minus 30 minutes to the axis of the centers of the punched holes.

3.4.1 Guide. A removable guide, either a stop or a pin, shall be mounted in the center of the left side of the pin register board (with the punches nearest the operator). The guide shall be adjustable to stop the edges of sheets being punched 14-3/4 and 15 inches from the centerline of the board. The guide shall be marked "29-1/2" and "30", respectively, to indicate its position. When a cammed stop is used, the markings shall be on the stop; when a pin is used, the markings shall be adjacent to the respective holes. The guide shall be secured to the board by a chain or braided wire covered with vinyl 8 inches in length and .047 in diameter or the guide shall have an alternate position whereby it may clear any material on the board.

3.5 Punches. The punches shall cut clean, free, round or rectangular holes without burrs or hanging punched-out material. The ends of rectangular holes may be rounded (semiconical ends). The punches shall punch cleanly through new aluminum offset, lithographic press plates, photographic film, and polyethylene terephthalate sheets 0.0075-inch thick. No additional aids, tools, or extensions shall be required to effect the punching. The punch pin and die shall be of A2 or O1 tool steel and shall be machined to tolerances of plus or minus 0.0005 inch. The die shall be 0.0005 inch larger than the pin. Punched-out material shall not build up so as to block the dies. Punches shall have four throat depths, 1-1/8, 5/16, 5/8 and 1/4 inch plus or minus 0.005 inch to the center of the hole. Gate pins or similar stops shall control the depth; the maximum depth may be controlled by the back of the throat wall. Gate pins shall clear the throat entirely when in the withdrawn position and shall fit into a hole drilled not less than .031 inch into the floor of the throat when in the inserted position. When gate pins are entirely removable, the gate pins shall be secured to the pin register board by a chain or similar device to prevent loss. Punches shall

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be individually operated so as to permit the operator to punch one hole at a time. Punch pins and levers shall remain in either the withdrawn (up) position or inserted (down) position by means of set screws, catches, or similar devices. Lowered pins shall return to the withdrawn position, thereby clearing the throat, when the lever is raised.

3.6 Board. Holes punched by the board shall be on 13.4375-inch plus or minus 0.005-inch centers. There shall be three rectangular holes, 3/16 by 3/8 inch. The middle hole shall have its long axis coincident with the centerline scribed on the working surface. The two outside holes shall have a common long axis at right angles to the middle hole which shall pass through the center of the middle hole.

3.7 Pin register bars. Pin register bars shall be of corrosion-resistant steel, 32 inches plus or minus 1/16 inch long, 1-1/2 inches plus or minus 1/16 inch wide, and 0.015 plus or minus 0.005 inch thick. The appropriate number and size of corrosion-resistant steel pins shall be spot welded along the bar, centered between the sides; the extreme pins shall be equidistant from the ends of the bar. The pins on the bars shall be 3/16-inch round or 3/16-inch square, spaced on 13.4375-inch plus or minus 0.005-inch centers. When the pins are square, the sides of the pins shall be parallel to the sides and ends of the bar. Pins shall project 3/16 inch plus or minus 1/32 inch above the surface of the bar.

3.8 Pin register plates. Pin register plates shall be of corrosion-resistant steel, 1 inch plus or minus 1/16 inch by 1-3/4 inches plus or minus 1/16 inch by 0.010 inch plus or minus 0.005 inch. A corrosion-resistant steel pin of the size appropriate to the type shall be spot welded off center of both axes (i.e., in one quadrant) to facilitate handling. The pins on the plates shall be 3/16-inch round or 3/16-inch square. The stud shall project 3/16 inch plus or minus 1/32 inch above the surface of the pin.

3.9 Identification marking. The pin register board and bar shall be identified in accordance with MIL-STD-130.

3.10 Workmanship. The pin register board, bars, and plates shall be free from defects that could impair their operation or serviceability. Surfaces shall not be scratched, warped, or dirty. All external edges and corners shall be rounded or beveled to minimize injury to personnel handling this equipment. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished product.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Examination. Each pin register board, bar, and plate shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Each pin register board shall be tested as specified in 4.4.2.1, 4.4.2.2 and 4.4.2.3. Failure to pass any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The pin register boards, bars, and plates shall be examined for the following defects:

101. Components missing, damaged, or not as specified.
102. Dimensions and tolerances not as specified.
103. Materials not as specified.
104. Construction and assembly incorrect or incomplete.
105. Working surface not as specified, or viewing area not as specified; does not transmit 80 percent light.
106. Poor fit at joints, seams, or edges, or misalinement.
107. Workmanship not as specified.

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4.4.2 Tests.

4.4.2.1 Distortion test. The pin register board shall be placed vertically on its edge on a solid surface and a load of not less than 94 pounds shall be layed across the top edge. The board shall then be rotated so that another edge is uppermost, and the same weight shall be applied to this edge. This procedure shall be repeated until the load has been applied to all edges. The board shall be placed in its normal operating position and a load of not less than 50 pounds shall be applied to the center of the board over an area not to exceed 20 square inches or a 5-inch-diameter circle. Any evidence of deformation remaining after these loads have been removed shall constitute failure of this test.

4.4.2.2 Operational test. Sheets or strips of photographic film, polyethylene terephalate plastic 0.0075 inch thick, and aluminum offset lithographic press plates shall be punched in the pin register board and then examined and measured. Failure to cut clean holes of the specified size and shape and in the specified locations without the use of tools, extensions, or other aids shall constitute failure of this test.

4.4.2.3 Line-up test. The line-up test shall be performed twice, once at each throat depth. A sheet of translucent plastic shall be placed on the board, against the stops, and punched. While the punches are in the lowered position, a line shall be drawn on the sheet coincident with the centerline on the working surface. The sheet shall then be turned over; the drawn line shall be alined with the centerline, and punched again. The sheet shall then be examined. Any evidence of repunching outside the original holes shall constitute failure of this test.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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108. Materials not as specified for levels A and B.
109. All components not preserved and packaged together as specified.
110. Strapping not zinc coated for level A.
111. Gross weight of fiberboard boxes exceeds the weight limitation of the box specification for level B.
112. Fiberboard boxes not waterproof sealed and taped as specified for level B.
113. Marking illegible, incomplete, or incorrect.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified.

5.1.1 Level A. Each pin register board, with pin register bars and pin register plates, shall be packaged together in a box conforming to PPP-B-636, W6c, style RSC. Cushioning shall be provided to prevent movement of the items within the box. After closure, the box shall be waterproof sealed, as specified for slotted style boxes, in accordance with the appendix to the box specification.

5.1.2 Level B. Each pin register board, with pin register bars and pin register plates, shall be packaged together in a box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125. Cushioning shall be provided to prevent movement of the items within the box. Closure shall be in accordance with the appendix to the box specification.

5.1.3 Level C. Each pin register board, with pin register bars and pin register plates, shall be packaged together in a container that will afford protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be level A, B, or C as specified.

5.2.1 Level A. The pin register boards with components, preserved and packaged as specified in 5.1 shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, style optional. The boxes shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc-coated.

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5.2.2 Level B. The pin register boards with components, preserved and packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style RSC or boxes conforming to PPP-B-640, class 2, style optional. The gross weight and size of the boxes shall not exceed the limitations of the applicable box specification. After closure, the boxes shall be waterproofed sealed with tape and reinforced with bands of tape in accordance with the appendix to the applicable box specification.

5.2.3 Level C. The pin register boards with components, preserved and packaged as specified in 5.1, shall be packed in containers that will assure carrier acceptance and safe delivery from the supply source to the initial destination at the lowest ratings of the carrier. However, containers as a minimum shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or purchase order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pin register board, pin register bar, and pin register plate are intended for the following uses:

(a) Punching cartographic scribing material and film to maintain accurate register in the preparation of color proofs.
(b) Punching photolithographic plates to maintain accurate registration of multicolor map separations when preparing offset plates.
(c) Facilitating registration of multicolor map separations when processing cartographic and photolithographic materials.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packaging and level of packing required (see 5.1 and 5.2). Level B preservation and packaging is intended to provide limited but economical protection and should be specified only when it is determined that pin register boards, bars, and plates will be held in covered storage approximately 1 year from date of initial preservation and packaging.
(c) Any special marking required (see 5.3).

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6.3 Figure 1 shows pin register board, bar, and plate which have been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other designs which conform to this specification. The use of this design does not relieve the supplier from complying with all requirements of this specification.

6.4 Type I. Type I pin register punch boards have been discontinued since they are no longer required.

Custodians:

Army - ME

User activities:

Army - GL

Preparing activity:

Army - ME

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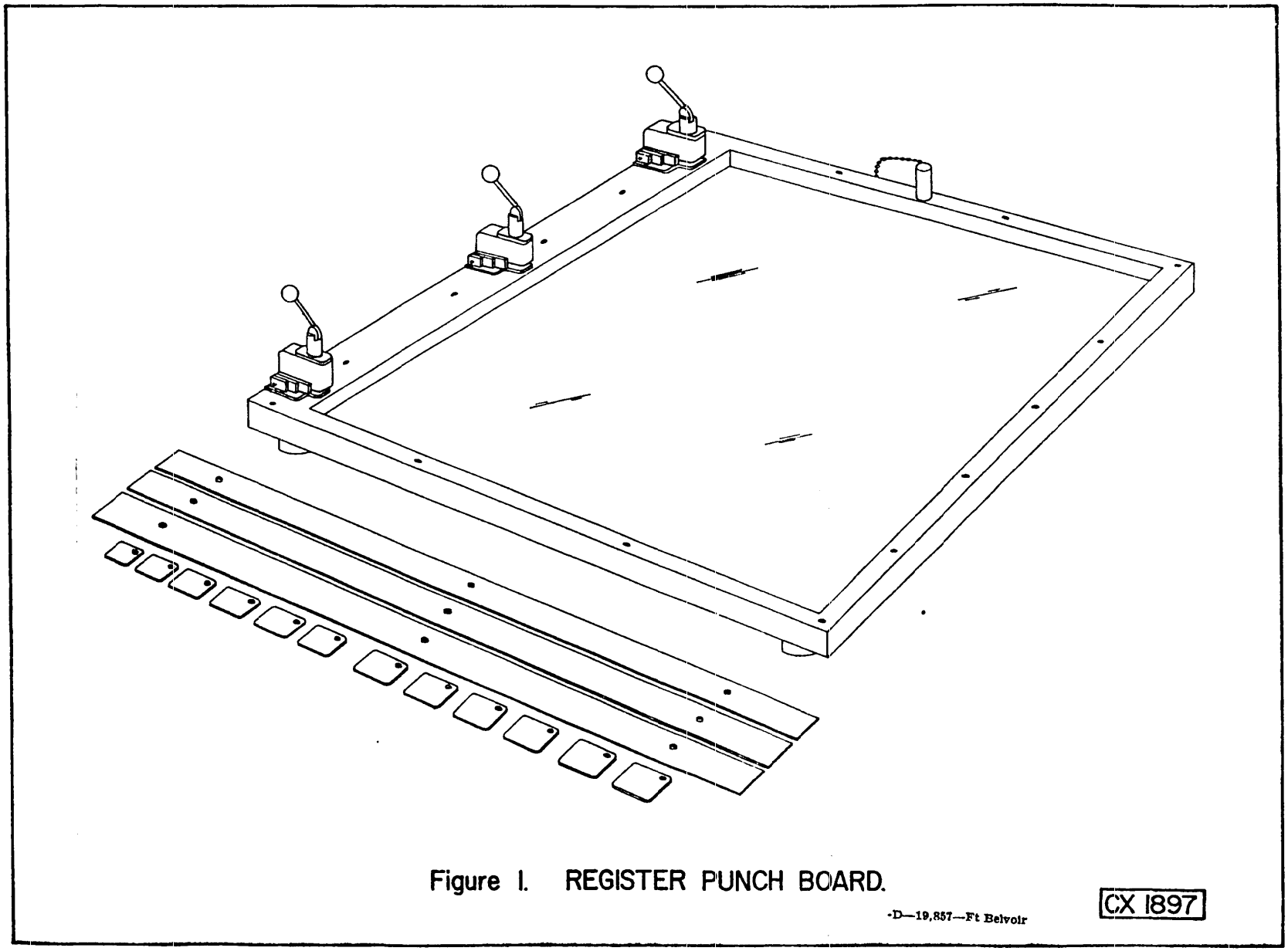


Figure 1. REGISTER PUNCH BOARD.

-D—19,857—Ft Belvoir

CX 1897

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-P-52455A(ME) Pin Register Punch Board, Pin Register Bars, and Pin Register Plates

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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